313357

12425

3 Hours / 70 Marks Seat No. | | | | | | | | |

Instructions : (1) All Questions are *compulsory.*

(2) Illustrate your answers with neat sketches wherever necessary.

(3) Figures to the right indicate full marks.

(4) Assume suitable data, if necessary.

(5) Mobile Phone, Pager and any other Electronic Communication devices are not permissible in Examination Hall.

Marks

1. Solve any FIVE : **10**

(a) Define : (1) Rubber (2) Elastomer

(b) State any four properties of Silicon Rubber.

(c) Define Synthetic Rubber. Give example.

(d) Define compounding of Rubber.

(e) State function and example of filler.

(f) State principle of rubber extrusion process.

(g) State significance of Resilience Test.

2. Solve any THREE : **12**

(a) Explain in brief Latex Technology.

(b) Give classification of Rubber with example.

(c) Describe preparation of Styrene Butadiene Rubber.

(d) State any four properties & applications of Neoprene Rubber.

3. Solve any THREE : **12**

(a) Elaborate – Latex Tapping Process.

(b) Differentiate between natural & synthetic rubber.

(c) State any four properties & applications of EPDM rubber.

(d) Explain manufacturing of Nitrile Butadiene Rubber.

4. Solve any THREE : **12**

(a) State the need and advantages of Thermoplastic Elastomer.

(b) State compound recipe for conveyor belt, state function of each additives.

(c) With neat sketch, explain working principle of compression moulding.

(d) State the need and effect of vulcanization on the rubber.

(e) Describe the test procedure for hardness test.

5. Solve any TWO : **12**

(a) Draw sketch of two roll mill and explain construction & working of two roll mill.

(b) Describe Rubber Injection Molding Process and name any four products produced from this process.

(c) Elaborate the following terminologies with respect to calendering process :

(1) Skimming (2) Fractioning (3) Topping

6. Solve any TWO : **12**

(a) With neat sketch describe construction and working of Banbury mixer.

(b) Enlist the type of vulcanization. Explain mechanism of Sulphur Vulcanization.

(c) Explain constructional features & working of ram extrusion for rubber with neat sketch.